

JINGZHANG ON TIME · A3 Booklet (v2.0)

v2.0 | 2026-08-14 | YuanYii

02 Site Analysis and Data Boundaries

All claims are reversible and reviewable. Drawings are generated on provisional constraint boundaries and do not constitute statutory control red lines. Official polygons, building census, FAR and real road networks are not yet published — all areas are provisional (official_boundary=false), buildings.geojson keeps status=unknown, and the demolition-retention-reconstruction proposal is a conceptual draft (confidence=low). Operating-diagram KPIs remain unknown until the first-service pilot: design blueprints are never presented as measured performance.

Verified context (public sources): the ~9 km corridor with the first 2.5 km / 16.8 ha phase open; Qinghuayuan Station as a century-old anchor; 37 universities along the corridor; TOD stations (Wudaokou / Qinghua East Road West / Dazhongsi); ~15 main-spine pedestrian gaps (EPSG:4548 measured basis carried over).

03 Three-Tier Working Framework

Regional study (43.6 km²) sets strategy and regional coordination; overall design (11.41 km²) settles land use, blue-green, mobility and urban form; key areas (368.4 ha) detail three districts. The layers are non-overlapping and seamless.

Overall concept: One Diagram, Nine Stations, One Spine. The heritage park becomes the operating-diagram main line; three stations (Departure Yard / Zero-Kilometre Station / Arrival-Transfer Station) carry the verify-translate-adopt responsibility chain; the Urban Operating Diagram is the governance core: every AI public service registers a published timetable — punctuality verifiable, changes noticed, suspensions announced, delays compensated, transfers connected.

Tier Area Scope
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Regional study 43.6 km ² Southern Haidian innovation corridor (5th Ring Road N / G6 E / Xizhimen Outer St S / Wanquanhe Rd W)
Overall design 11.41 km ² 1–2 km urban area around the heritage park
Key areas 368.4 ha Zhongzhi Park 192.1 + Origin Community 104.3 + Dazhongsi 72.0

04 Global Cases and Regional Coordination

Six benchmarks contribute mechanisms only, with explicit no-copy boundaries (no land regimes, regulatory powers, capital conditions or building forms): Mission Bay (capital+research dual channel → Zero-Kilometre Station), Silicon Roundabout (brownfield × creative industry → Arrival-Transfer Station), MaRS (three-party governance → operating community), Cornell Tech (campus-enterprise sandbox → first-service pilot), Kendall Sq. (walkability grading → five-band spine), Punggol (system interconnection → window maintenance). Regional coordination (Zhongguancun service wing / Xiaoyuehe scenario wing / Future Science City / E-Town) remains open_interface_no_commitment.

05 Overall Design: Land Use and Renewal

Four land categories total 1141.3 ha = 11.41 km², non-overlapping: R&D innovation 311.9 ha (27.3%), green open 270.8 ha (23.7%), industry-commerce 309.4 ha (27.1%), talent housing 249.2 ha (21.8%). Renewal follows a demolish-retain-reconstruct conceptual draft (confidence=low, pending building census): Origin Community — retain and restore (Qinghuayuan Station, university buildings); Zhongzhi Park — retrofit (riverside industrial stock); Dazhongsi — demolition discussion (illegal structures, pending engineering survey); whole line — reserve small new facilities (signal posts, timetable boards, TOD links).

06 Mobility, Transit and Municipal Infrastructure

TOD stitching at Wudaokou / Qinghua East Road West / Dazhongsi (15-minute rail-walk circles; elevated links under municipal feasibility review). Main-spine five-band slow mobility; autonomous shuttle micro-circulation as a registered concept service. Quantified scenarios are synthetic pending field verification: green 300 m coverage 85.6%, TOD 500 m union 20.6%, ~15 greenway gaps (EPSG:4548 basis). Transfer protocol: cross-district handovers follow transfer times, receipts require human sign-off, delays must never silently skip a transfer (KPI-5). Municipal: rooftop PV + micro-grid exploration; edge computing micro-stations on reversible plug-rails; data minimization with 7-day destruction; the weekly window doubles as a whole-line equipment and energy inspection slot.

07 Blue-Green Space and Urban Form

One axis, two rivers, multiple corridors, a hundred parks: the heritage park as the north-south ecological spine, with Qinghe and Xiaoyuehe; continuous greenways connect 12 communities and universities. Green patches total 3.55 km² (31.1%); the 9.5 km green ridge wind corridor estimates 0.8–1.5 °C cooling (synthetic, pending CFD). Qinghe low-carbon waterfront runs multi-modal ecological sensing (water quality / bird calls / micro-climate) under the operating diagram and window maintenance list. Material language: century-old red brick (heritage), Zhongguancun grey aluminium (industry), AI transparent glass (future landmarks); signal posts and timetable boards form a

unified recognizable kit (navy base + signal three-colour semantics). Cultural timeline: railway 1905–1909 (Zhan Tianyou's switchback → self-reliance), Zhongguancun 1980s–2010s (electronics street → national innovation zone → pioneering), AI 2010s–future (open source, foundation models, agent collaboration → open co-creation).

08 Key Areas: Three Stations

Zhongzhi Park · Departure Yard (192.1 ha, ZJ-01). Garden-style full-stack innovation district. The pre-departure marshalling function: every service completes paired testing (AI on / AI off) and baseline verification here before receiving its "punctuality seal" on the diagram. 100-metre window: four-state section at the marshalling-yard front plaza (marshalling / trial / delay / suspension). Anchors: T-01 registration, signal posts SP-01/02.

Origin Community · Zero-Kilometre Station (104.3 ha, ZJ-02). University-adjacent translation community. Qinghuayuan Station adaptive reuse as the Timetable Hall (public service-schedule interface); stitching of Qinghua East Road West / Wudaokou gaps; low-cost open collaboration spaces in converted industrial buildings. The first-service pilot cabin (T-07) anchors here — the full 90-day first-service loop. 100-metre window: "first departure ceremony" sequence (arrival — timetable hall — transfer desk — staffed service band). Anchors: T-04 transfer desk, T-07 pilot cabin.

Dazhongsi · Arrival-Transfer Station (72.0 ha, ZJ-03). Urban smart-economy district. TOD integration and a four-quadrant elevated walkway (under municipal feasibility review); the arrival plaza kit (signal post + punctuality board + compensation kiosk); the Diagram Yearbook Hall (T-08) anchors here — annual review and failure archive made public. 100-metre window: four-state section at the arrival plaza.

Intermediate Stops (MS-01~06). Light stops along the spine (every ~1.3 km): Qinghe/North 5th Ring, Xueqing Road, Wudaokou, Beitapingzhuang, Xueyuan South Road, Dazhongsi North, southern terminus — anchored at university interfaces, community nodes and gap-stitching units (6 of 15 gaps upgraded). Each stop carries four light facilities: timetable information point (T-02 extension), signal post (co-located with SP), traveler rest (seating/drinking water/shade/bike-share), staffed service window (inquiry/guide/daily needs — no AI equivalent path). Functions: high-frequency daily stops, cross-district transfer handover points (T-04 extension, human-signed receipts), maintenance-window stop points (co-located with 6 window-plaza candidates). Access tier L1 filing; responsible roles R-STATION-OPS / R-COMMUNITY-CARE.

All three windows use the fixed five-band section order (timetable band — accessible band — slow-mobility band — staffed-service band — signal-post sequence), unchanged when AI is off.

All three stations and six intermediate stops sit on the nine geographic layers; the governance assertions are set out below.

Governance Attribute Assertions (Tier-1 requirement)

[data:geometry/key_areas.geojson#KA-001] [data:geometry/phasing.geojson#PH-01]

Governance objects passing the mechanism triple-check (input / decision logic / state-and-stop conditions) must carry attribute assertions on geographic layers; otherwise they are not counted in the governance tier. All 9 geojson layers assert zone_id / raci / gate / status:

Layer | zone_id | raci | gate | status | Assertion

|---|---|---|---|---|---

site_boundary.geojson | ZN-PUB | R-DESIGN | G0 | active | Overall boundary (provisional, official_boundary=false)

key_areas.geojson | ZN-PUB/ZN-KA | R-DESIGN/R-PLAN | G1 | active | Three key areas (KA-001/002/003)

land_use.geojson | ZN-KA | R-PLAN | G1 | active | Four land-use classes, full coverage

green_space.geojson | ZN-PUB | R-GREEN | G1 | active | 10 green patches, 3.55 km²

public_space.geojson | ZN-PUB | R-COMMUNITY-CARE | G2 | active | Public-space nodes

roads.geojson | ZN-KA | R-MOBILITY | G2 | active | Spine 5-band & slow-traffic concept lines

buildings.geojson | ZN-KA | R-PLAN | G3 | unknown | Building survey unpublished — status=unknown registered

constraints.geojson | ZN-PUB | R-DESIGN | G3 | active | Empty constraint set (official constraints unpublished)

phasing.geojson | ZN-KA | R-DELIVERY | G1 | active | Six renewal projects phased (PH-01)

Rule: any layer with status=unknown is not a basis for governance effect; objects below gate G1 are not registered on the operating diagram (linked to T-01); raci roles map one-to-one to scenario-card responsibility fields (R-STATION-OPS / R-COMMUNITY-CARE / R-DELIVERY). Since v2 the in-text assertion table and geojson properties are cross-checked bidirectionally; layers missing any element do not enter mechanism triple-check.

09 AI Innovation Ecosystem and 8 Scenario Cards

Eight scenario cards serve the operating-diagram loop only (deliberately restrained, avoiding the homogeneous scenario sets of high-scoring proposals). Each card carries eight structured fields (space / access tier / TRL / AI-off path / stop condition / transfer / delay threshold / responsible role) enforced by the diagram checker — a card missing any field is not counted as service coverage.

ID | Scenario | Space | Tier | TRL | Mechanism in one line

|---|---|---|---|---|---|

T-01 | Diagram Registration Desk | all stations | L1 | 7-8 | No timetable, no departure — unregistered services do not run

T-02 | Punctuality Board | station halls | L1 | N/A | Monthly punctuality published; delay over threshold turns amber + compensation notice

T-03 | Window Plaza | spine | L1 | N/A | Weekly fixed maintenance window: AI off line-wide, public space returns to no-AI state

T-04 | Transfer Desk | district joints | L2 | 6-7 | Cross-district handovers follow transfer times; receipts signed by humans

T-05 | Delay Compensation Kiosk | all stations | L1 | N/A | Delay over threshold auto-opens human alternative / free equivalent / appeal

T-06 | Signal Post | spine + stations | L1 | N/A | Green / amber / red public status code

T-07 | First-Service Pilot Cabin | Origin 100m window | L2 | 5-6 | Full 90-day first-service loop (register-trial-assess-release/suspend)

T-08 | Diagram Yearbook Hall | Dazhongsi | L1 | N/A | Annual review public, failures and delays archived — the diagram does not lie

Access tiers: L1 open display (5 cards; filing system, immediate disconnect on risk); L2 controlled testing (2 cards; joint approval + ethics review + staffed control); L3 sandbox reserved (whitelist + physical isolation + enforced data destruction).

Seven user profiles: open-source developers / startups / enterprise visitors / residents / university members / elderly·disabled·no-phone users (most-disadvantaged: priority rehearsal + veto) / night workers (protected timetable). Averages must never mask single-group failure — group results are reported separately.

Three landmarks: the JZ AI Meridian Coordinate Tower (heritage park × Qinghua East Road), the Zero-Kilometre Timetable Hall (adaptive reuse), the Diagram Yearbook Hall (Dazhongsi) — commemorating contribution and record, not spectacle.

10 Operating Diagram Governance Protocol and Open Operation

Urban Operating Diagram (v1.0). Every AI public service registers its spatio-temporal schedule (service ID / interval / window / punctuality definition / delay thresholds / change notice / suspension notice / transfer / AI-off path / stop condition). Publicly queryable, machine-readable (timetable.schema.json), machine-verifiable (check_timetable.js; 10 fixtures: 3 accepted, 7 rejected).

Three-signal display. Green = punctual; amber = delayed with human takeover; red = suspended, reverting to the no-AI baseline. Signal posts make governance state spatially legible.

Punctuality KPIs. Punctual = completed on schedule without human takeover. Six KPIs (punctuality / delay / suspension fulfilment / window execution / transfer connection / compensation fulfilment) are sourced from the diagram log, published monthly, archived annually.

Weekly window. Every Tuesday 06:00–12:00: AI off line-wide for maintenance (equipment / model / data / interface), public space returns to no-AI state — periodic AI exit becomes civic rhythm, not emergency. Execution enters KPI-4.

Transfer protocol. Cross-district handovers follow transfer times; receipts require human sign-off; delays never silently skip a transfer; no receipt, no operation.

Gates G0–G4. Registration → baseline verification → controlled trial → punctuality assessment → release/suspend. Each gate has inputs, judgement and stop actions; years and hype never substitute for evidence gates. Suspension is a qualified outcome (falsify-first).

Paired testing (AI on / AI off). Same task, both states; equivalence gap (completion / time / accessibility / appeal) is mandatory evidence for registration; denominators include failures.

Operating community (five parties). District authority (lead + authorization), park operator (execution), university alliance (resources), community representatives (veto), independent audit (review). No single party decides everything. Legal basis: Urban Renewal Regulation + Generative AI Interim Measures Art. 14/15 + Zhongguancun sandbox framework. Five-fund firewall (public base / research commission / public-interest commission / time-limited sponsorship / operating revenue); three ledgers (capex / opex / exit reserve) required before any G3 pilot. Civic Value Protocol: 10–20% of corridor commercial computing-node and scenario-test revenue studied for injection into the community public-value fund (voluntary CSR charter, ratio and rules to be determined by authorities and community representatives).

11 Renewal Projects and Phased Implementation

Six renewal projects run inside the diagram Proof-Mile loop (verify small first, scale on evidence): JZ-01 gap stitching (near term, connectivity KPI), JZ-02 signal-post + timetable-board samples (near term, online rate >95%), JZ-03 factory conversion (mid term, occupancy >80%), JZ-04 elevated links (mid term, flow KPI), JZ-05 edge computing nodes (long term, PUE <1.2), JZ-06 window maintenance system (near term, 100% execution).

90-day first service. D0–30 diagram registration and samples; D31–60 trial operation of the demonstration interval; D61–90 punctuality assessment and G3 decision. Cost estimate 405–820k RMB (BOM basis, not a budget commitment); recruitment targets (10 most-disadvantaged volunteers / 30 general testers / 6 interns) stated as not-yet-recruited; resource release tied to gates (10/25/25/25/15%). Suspension is a qualified outcome — the first phase succeeds by completing three falsifiable judgements, not by launching three AIs.

Execution Evidence and Log Baseline

[data:geometry/phasing.geojson#PH-01] [metric:punctuality_rate]

Verifiable current evidence (v2): Phase-1 of the Jing-Zhang Heritage Park corridor (2.5 km / 16.8 ha) is already in operation (publicly verifiable) — the physical basis and ridership anchor of the diagram main line; the 15 current spine gaps are EPSG:4548 measured (registered as baseline); 37 universities and the TOD skeleton (Wudaokou / East Qinghua Rd W. / Dazhongsi) are publicly verifiable.

Log contract (v2, code-level): visual/assets/logs_schema.json defines 6 event types (registration / trial / on-time / delay / suspension / compensation) with fields (service_id / time / stop / status code / human-takeover flag / evidence hash), isomorphic with check_timetable.js — logs start at pilot; KPIs use logs as the single source of truth; pre-pilot stays unknown, no fabrication (v1 commitment kept).

Pre-pilot baseline registry (v2): visual/assets/baseline_registry.json registers 3 baselines — operating corridor (2.5km/16.8ha), current gaps (15), TOD stop circles (3) — each with source and measurement basis; the pilot 90-day D61-D90 punctuality gate compares directly against them.

Evidence layering (v2): measured current (existing) → in-package recalculation (existing) → operation logs (post-pilot, contract first) → official-data recalculation (after release). Four layers, no mixing.

12 Indicator System and Compliance Matrix

All indicators trace to metrics.json; spatial indicators use the EPSG:4548 measured basis; diagram KPIs source from the diagram log and remain unknown until pilot (never disguised); synthetic items are marked and recalculated after official controls are published. Three-way separation: measured / package-recomputed / unknown-as-state-evidence.

Standard coverage 9/9: official announcement / agent taskbook / urban-design measures / control-detailed-planning / national land-use classification / architectural design depth / generative-AI interim measures / barrier-free law / elderly smart-tech plan.

Compliance separation (four columns). Statutory minimums (GenAI Measures Art. 14/15, Barrier-Free Law Art. 39, State Council 2020-45) are never waived by "design recommendation"; self-set standards (5-working-day appeals / 48h publication / L0 no-device baseline / 7-day retention) are labelled (self-set); each row states its spatial-operational consequence. Reading voluntarily adopted standards as universal legal obligations is itself misleading.

13 Risk, Copyright and Compliance

Ten risks with six attributes each (probability / impact / responsible role / trigger / stop-loss / recovery evidence); eight hard-stop triggers (data breach, takeover failure, public-safety equipment failure, privacy incident, supplier exit without equivalent, unsegregated appeals, missing equivalent path, three consecutive missed windows) — any hit means red-light suspension. Rollback pack of nine (stable version / configuration / data schema / manual SOP / on-site signage / public notice / deletion verification / responsible person / recovery checklist).

Boundaries: provisional geometry (official_boundary=false), whole-package recalculation on official data (no partial fixes); synthetic tabletop for simulation; COMMUNITY-DISPLAY-ONLY licence; concept proposals do not constitute administrative conclusions; SHA-256 copyright ledger; data gaps stated, not filled.

14 Drawing Index and References

This booklet is the display layer; all conclusions defer to the structured data layer (GeoJSON → metrics.json → compliance_matrix.json → this booklet). Drawings: overall design map (original basemap + diagram line / stations / signal posts), location analysis map (81-point coordinate annotation), operating-diagram key visual, three four-state sections, spine five-band section, station plans, diagram plaza concept. References: brief/public-brief.md (primary entry), data/source_registry.json, data/processed/agent_fact_pack.md; background materials (railway archives, Haidian statistics, W3C, WHO) serve as mechanism evidence only.

Drawings are generated on provisional constraint boundaries and do not constitute statutory red lines. All geometry, indicators and data are conceptual recommendations pending professional re-verification after official red lines are published.